Speaker Information

[Name] Yoko Yazaki-Sugiyama

[Institute] OIST Graduate University

[Position] Professor

[Email Address] yazaki-sugiyama@oist.jp

[Lab / Personal Website]
https://www.oist.jp/research/research-units/nmcpu

Talk Information

Learning to communicate by listening and vocalizing during development

Like human babies lean to speak, songbirds learn to sing through vocal communication with a tutor intensively during developmental critical periods. Juvenile zebra finches, a premier model of songbirds, first listen and memorize tutor's songs (TS) in the auditory learning phase and then match their vocals to the memorized TS in the following sensorimotor learning phase. We found a subset of neurons in the zebra finch higher auditory cortex, the caudomedial nidopallium (NCM) acquire selective responsiveness to TS after learning from it, suggesting TS memories form in NCM. We further found axonal projections in the song premotor area in the "song system", HVC from the NCM neurons responsive to playback of TS in juveniles, but not in adults. Ablating TS-responsive NCM neurons in juveniles, but not in adults, disrupts song learning. The density of axonal projections in HVC from TS-responsive NCM neurons negatively correlated with the level of song crystallization. Our studies have demonstrated neuronal circuits from/to the zebra finch higher auditory cortex, NCM, regulate efficient song memorization by vocal communications with adult tutors and auditory-memory guided song learning in well-orchestrated developmental auditory and sensorimotor learning period.

Research Overview (for collaboration discussion)

[Research Interests] Neuronal circuit development, Plasticity, Learning, Social behavior

[Current Research Projects] We have been investigating the neuronal mechanism underlying songbirds (zebra finches) to form memories and to learn to sing from vocal communication

[Techniques] Electrophysiology, Behavior, Viral vector, Molecular biology

[Aspects or techniques of interest for future collaboration]

Computational approach, Molecular biological /Biochemical approach